

Elicited Behaviors and Simple Learning Mechanisms

Learning and Behavior (PSY/NSC 3320)

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Baylor University

“elicit” = “to draw out or bring forth”

Elicited Behaviors – behavior drawn out by stimulus

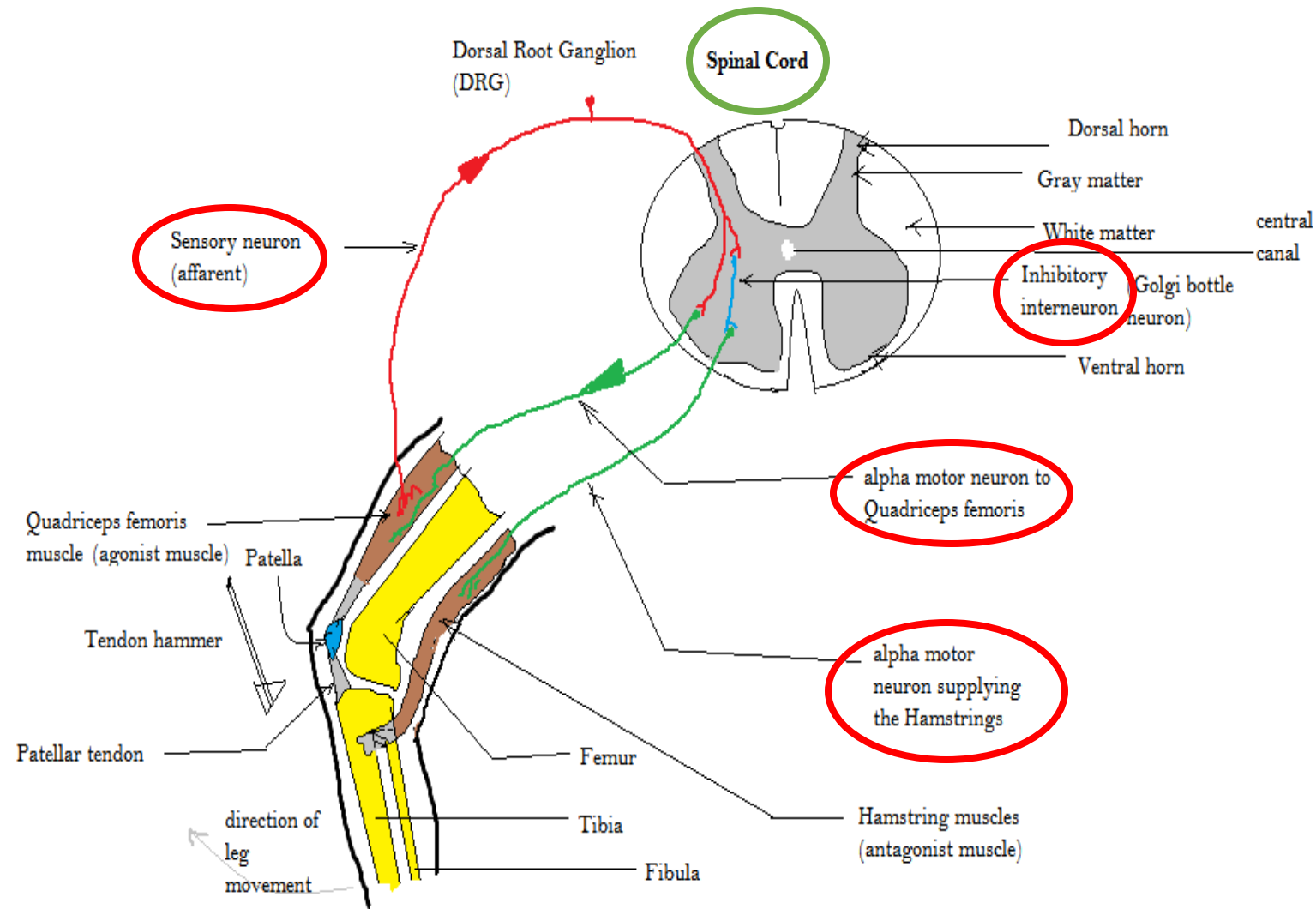
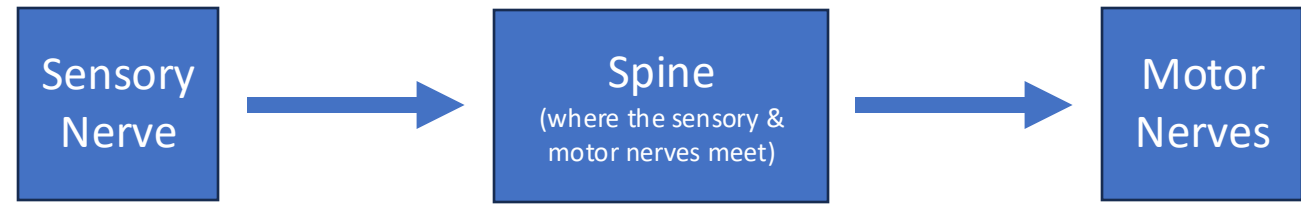
Elicited Behaviors

- **Reflexes** – automatic response to a stimulus
- **Fixed Action Patterns** – fixed complex response (sequence of actions) to a stimulus

Elicited Behaviors

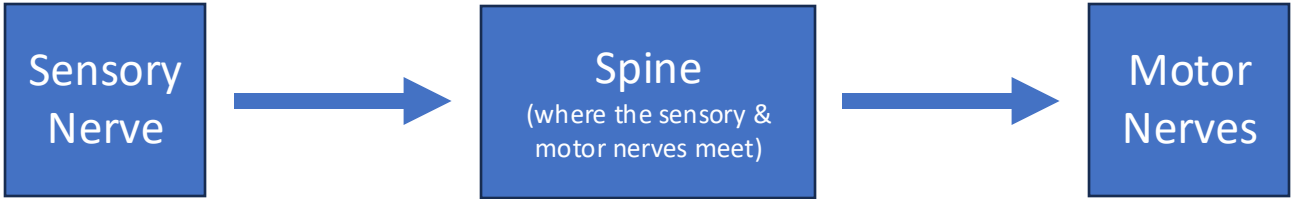
Reflexes – automatic response to a stimulus

- Reflex Arcs – *spine* – simple, in-born reflexes



Sensory nerve...

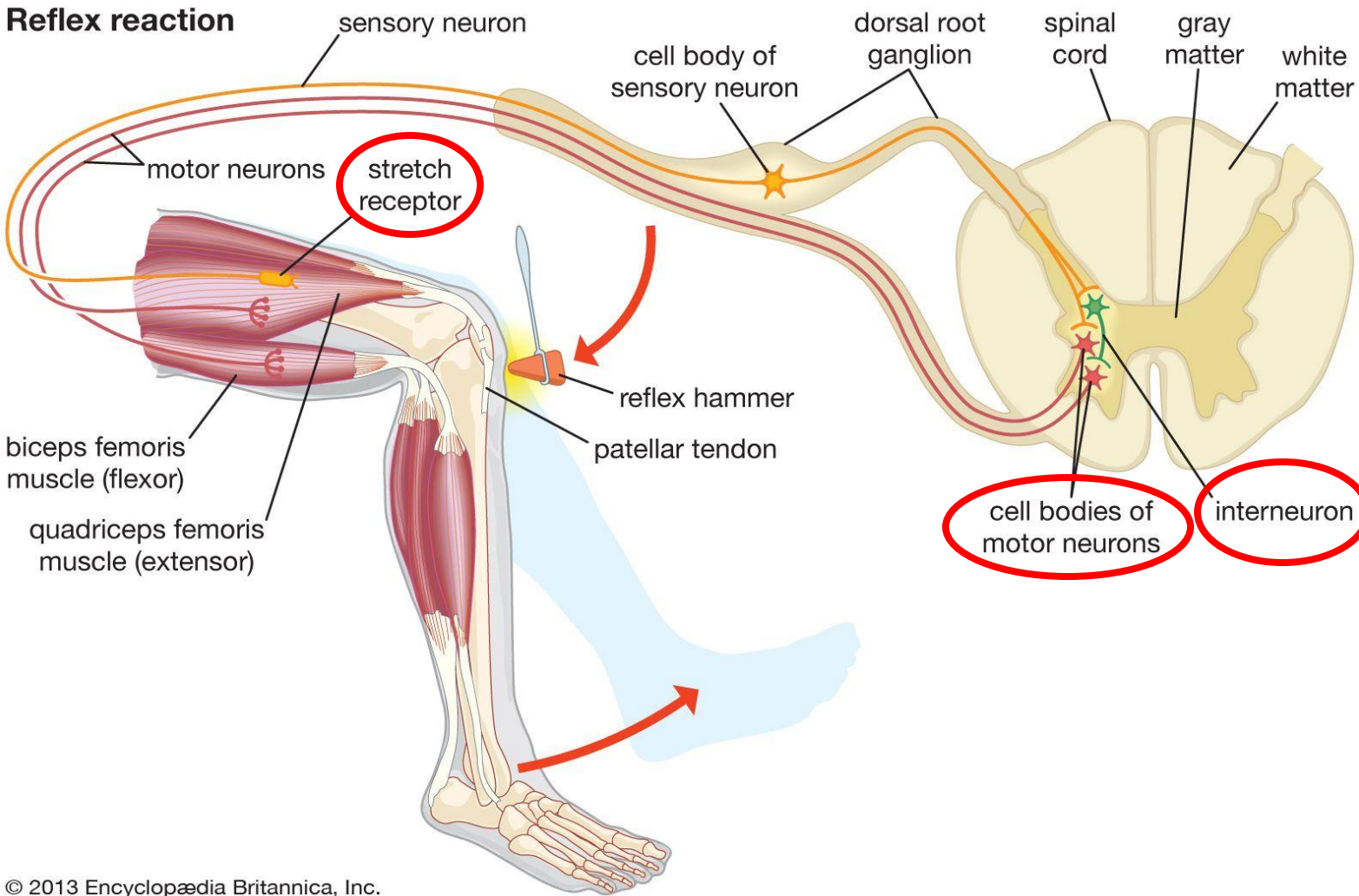
Activates quadricep motor neuron +
Activates interneuron → Inhibits hamstring neuron

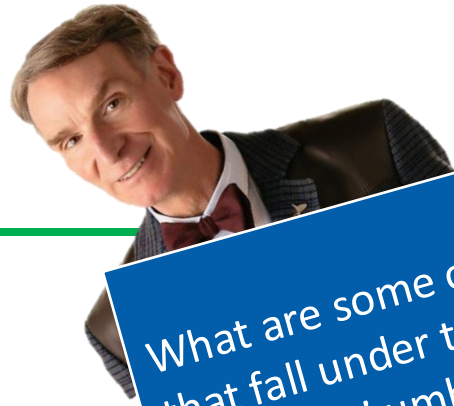


Elicited Behaviors

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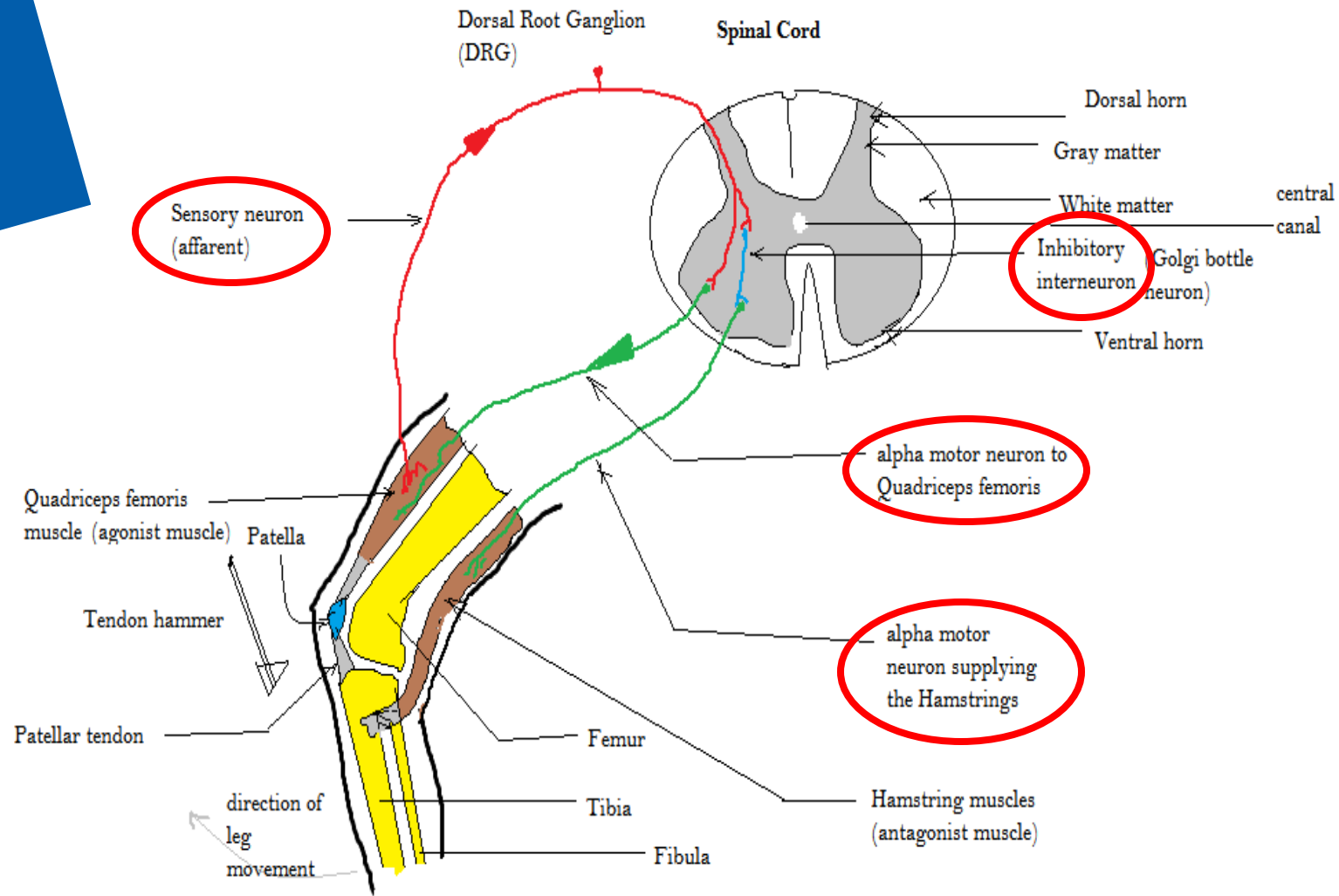


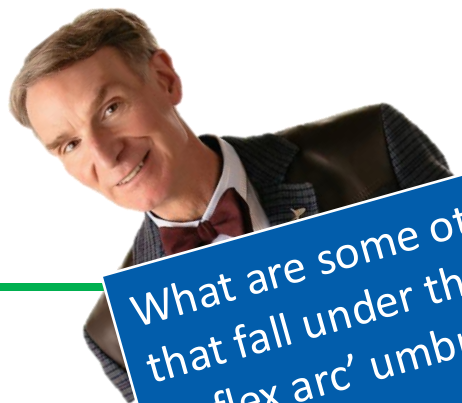
What are some other reflexes that fall under the common 'reflex arc' umbrella?

Elicited Behaviors

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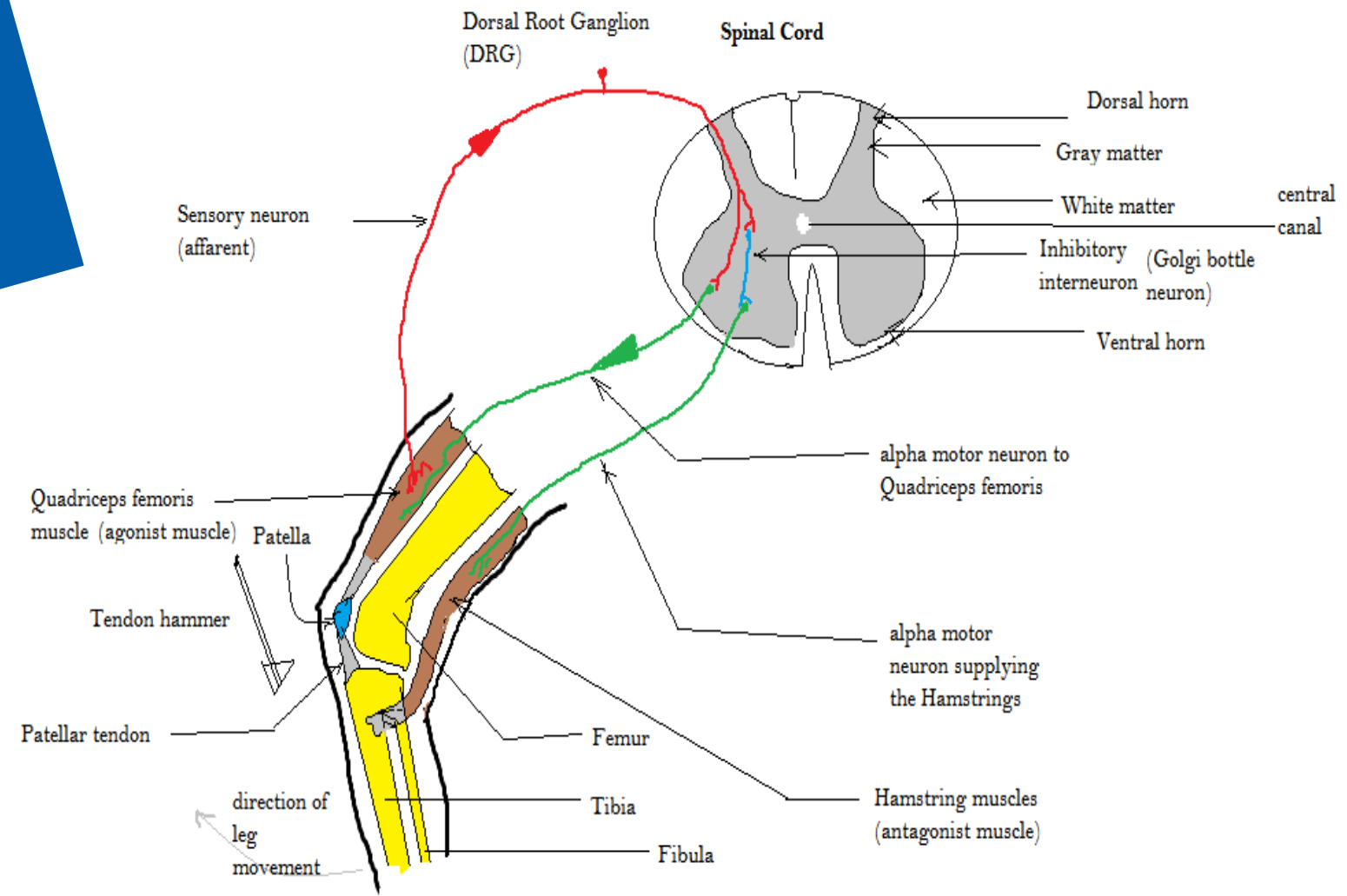
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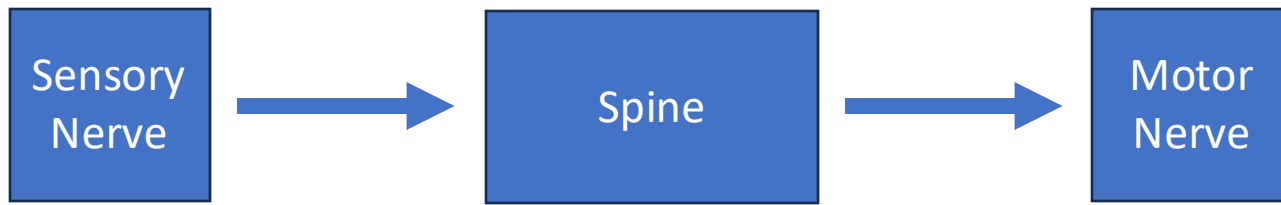
Patellar (pictured here)
Palmar
Blinking

Elicite

Reflexes – automatic response to a stimulus

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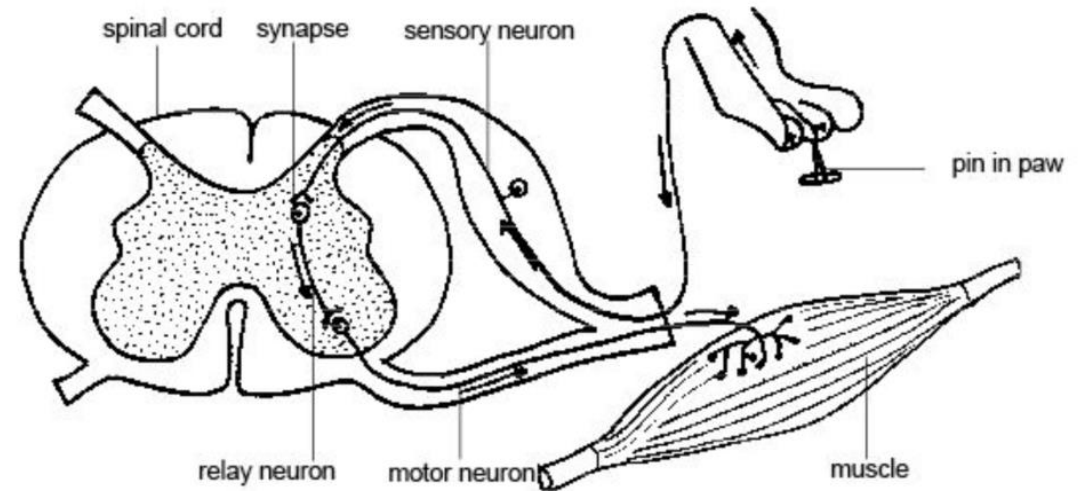
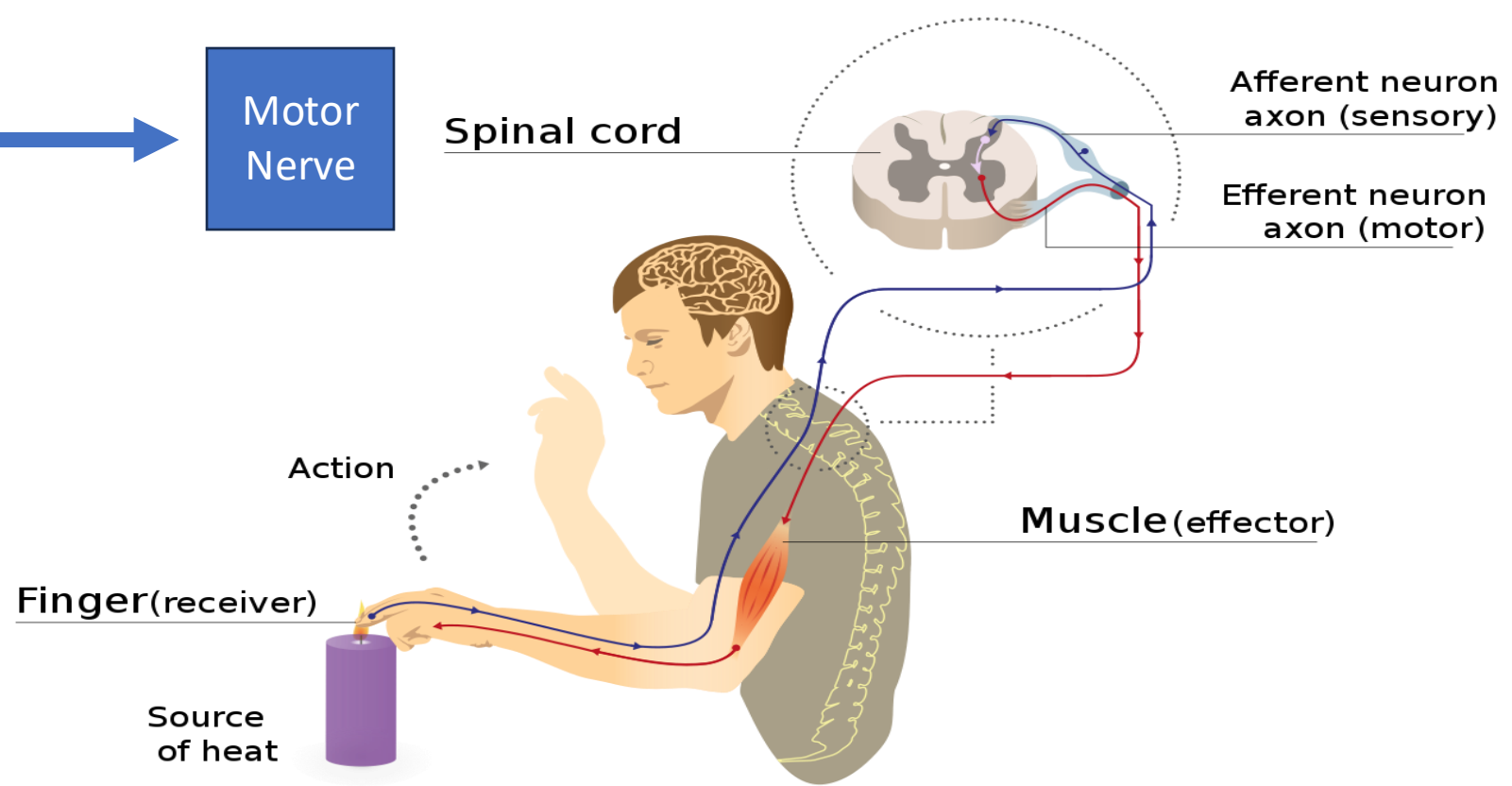




Elicited Behaviors

Reflexes – automatic response to a stimulus

- Reflex Arcs – *spine* – simple, in-born reflexes
- Flexion Response – *spine* – auto-recoil from hot/sharp object (*step on nail, don't 'feel' till after*)

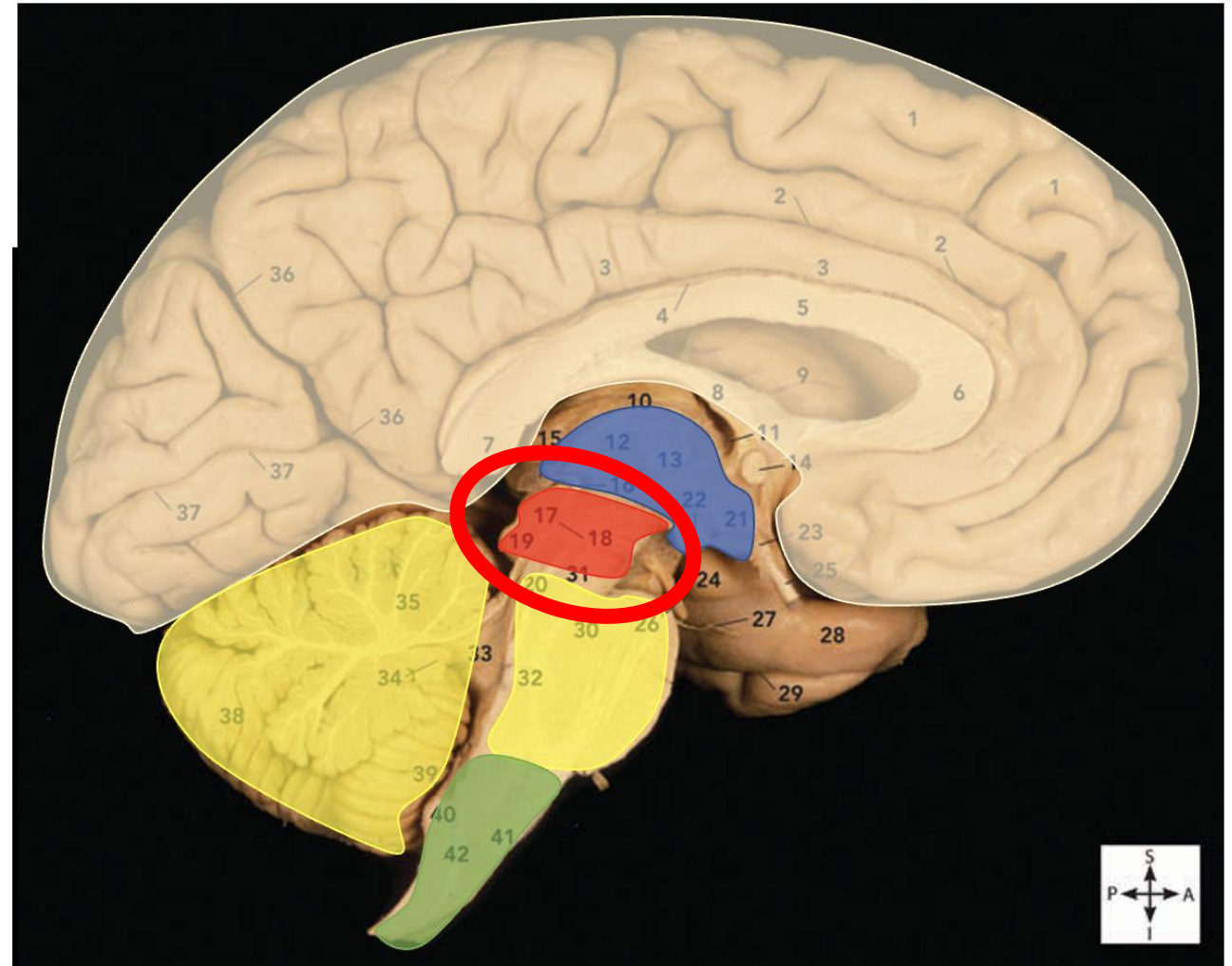


Sensory nerve in Ear or Eye → Midbrain → Shoulder/Neck muscles

Elicited Behaviors

Reflexes – automatic response to a stimulus

- Reflex Arcs – *spine* – simple, in-born reflexes
- Flexion Response – *spine* – auto-recoil pain stim
- Orienting Response – *Midbrain* – position body toward stimulus (see it, gain more info)

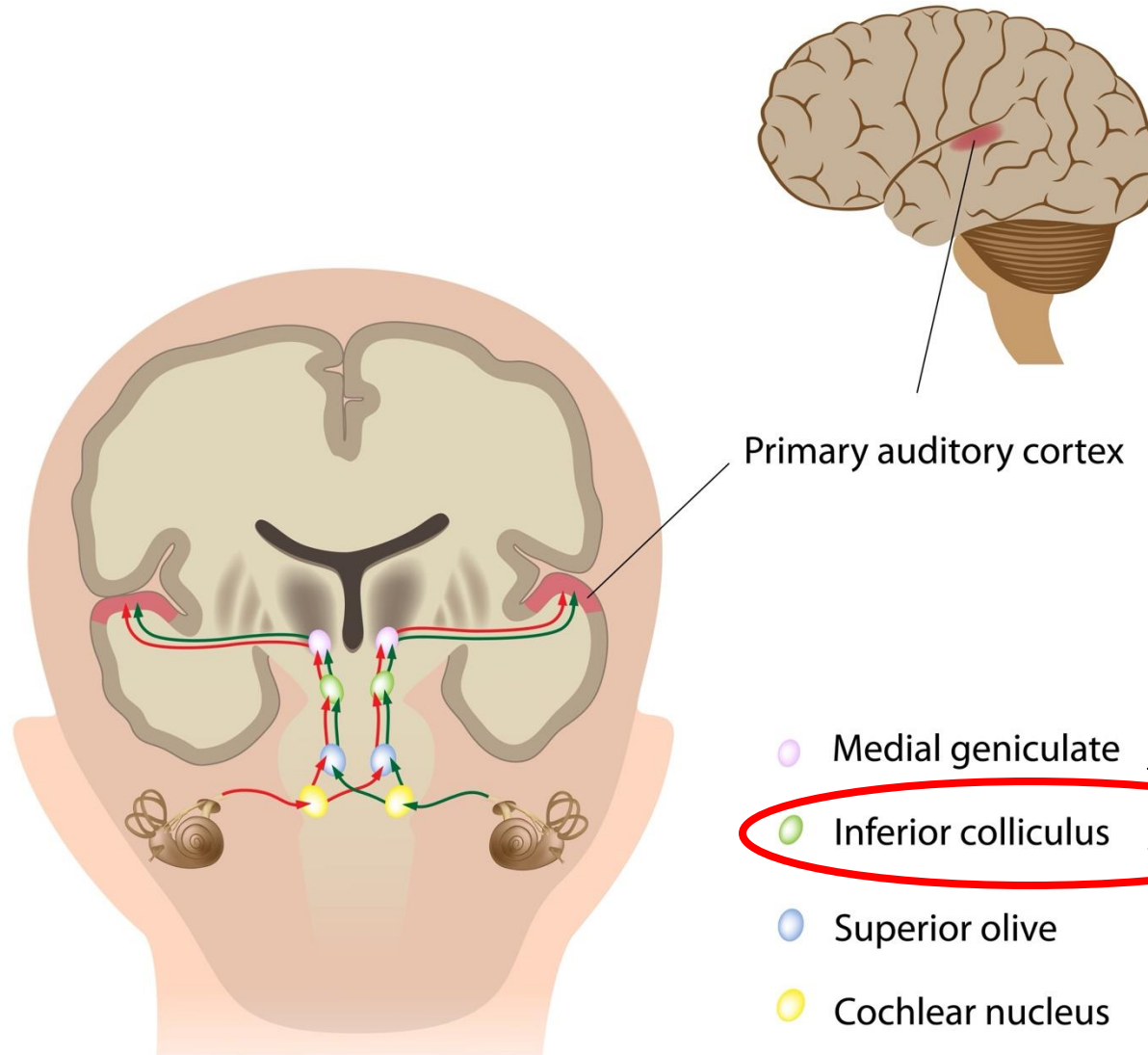


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Reflexes – automatic response to a stimulus

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- Flexion Response – *spine* – auto-recoil pain stim
- Orienting Response – *Midbrain* – position body toward sudden visual (typically a moving object) or auditory (typically a salient noise) stimulus (see it, gain more info)



Medial geniculate send info to STG

Inferior colliculus look at it

Superior olive

where'd it come from?

Cochlear nucleus

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- Startle Response – *Pons+Medulla* – defensive, whole-body reaction to unexpected stimulus



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Involves the hindbrain (Pons & Medulla) activating ancient fight/flight systems



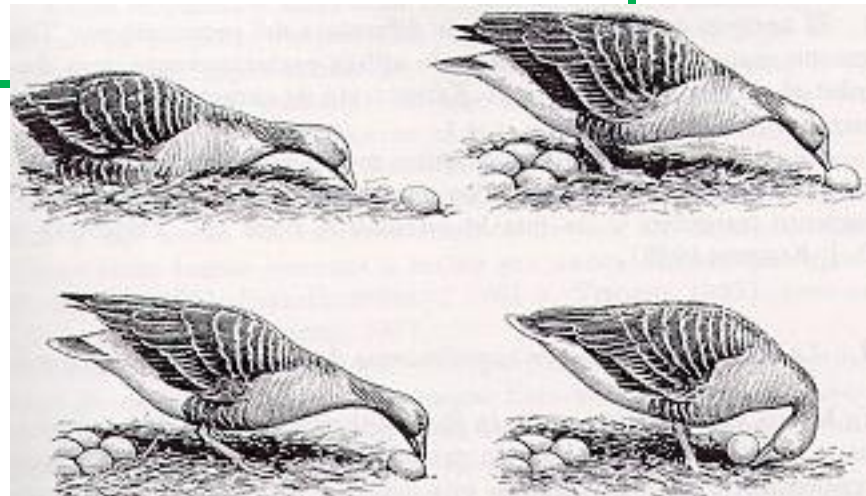
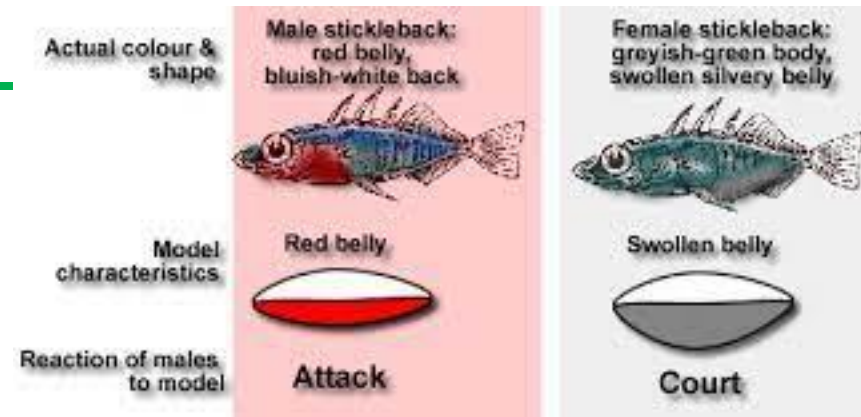
Elicited Behaviors

- Reflexes
- **Fixed Action Patterns** – fixed complex response (sequence of actions) to a stimulus

Elicited Behaviors

Fixed Action Patterns – fixed complex response to a stimulus

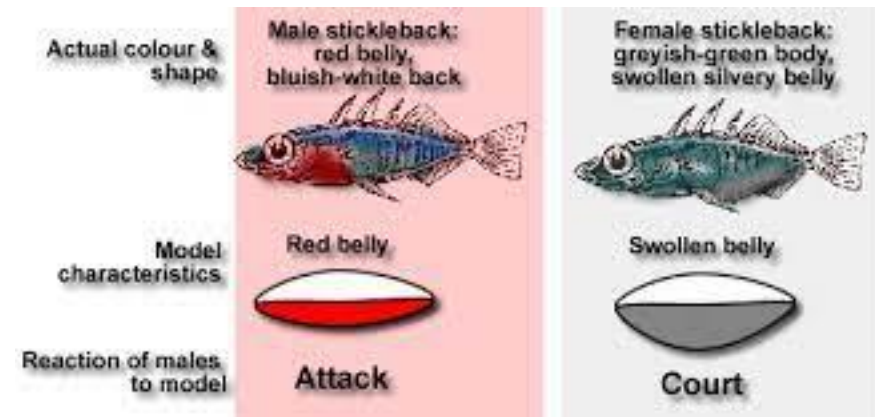
- Dog before play
- Geese rolling eggs into nest
- Stickleback fish aggression/courting
- Deer running zig-zags to avoid predators



Elicited Behaviors

Fixed Action Patterns – fixed complex response to a stimulus

- Stickleback fish aggression/courting
 - See another male's red belly → attack other male
 - Covered red belly on another male → no attack
 - **Simple stimulus → Fixed complex response**



Elicited Behaviors

Fixed Action Patterns – fixed complex response to a stimulus

- Sign Stimulus (or “releaser”) – sensory trigger for fixed action pattern

Elicited Behaviors

Fixed Action Patterns – fixed complex response to a stimulus

- Sign Stimulus (or “releaser”) – sensory trigger for fixed action pattern

Eleven characteristics of fixed action patterns:

1. Triggered by external stimuli
2. Composed of invariant sequence of actions
3. Not learned by individual organisms
4. Inherited
5. Performed by every member of a species
6. Do not vary between individuals in a species (stereotypical)
7. More complex than reflexes
8. Can be inappropriately triggered
9. Lack conscious purpose, can be imperfect
10. Threshold for triggering does not depend on environment
11. Individual actions share common triggers that differ from triggers of other patterns

Elicited Behaviors

Fixed Action Patterns – fixed complex response to a stimulus

- Sign Stimulus (or “releaser”) – sensory stimulus for fixed action pattern
- Characteristics:
 - Externally triggered
 - Complex
 - Innate
 - Unconscious
 - Stereotypic
 - Invariant

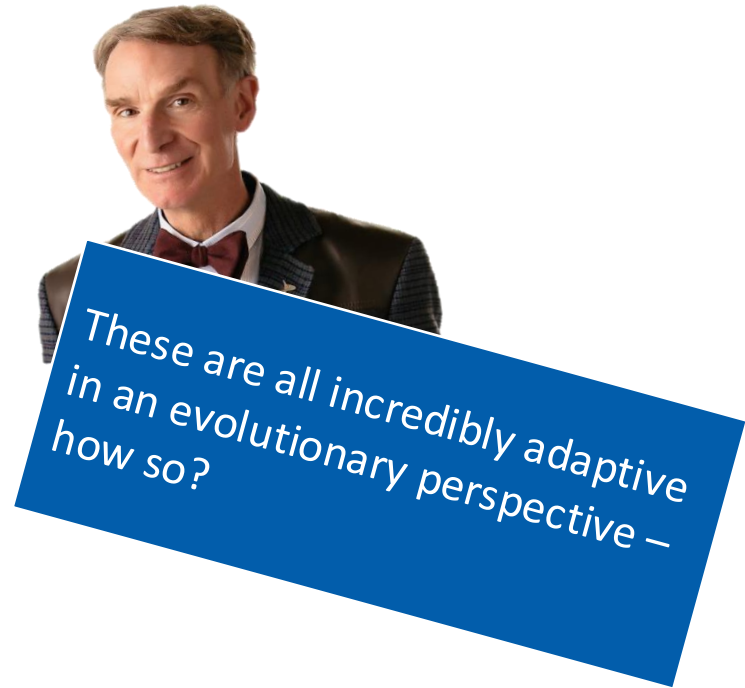
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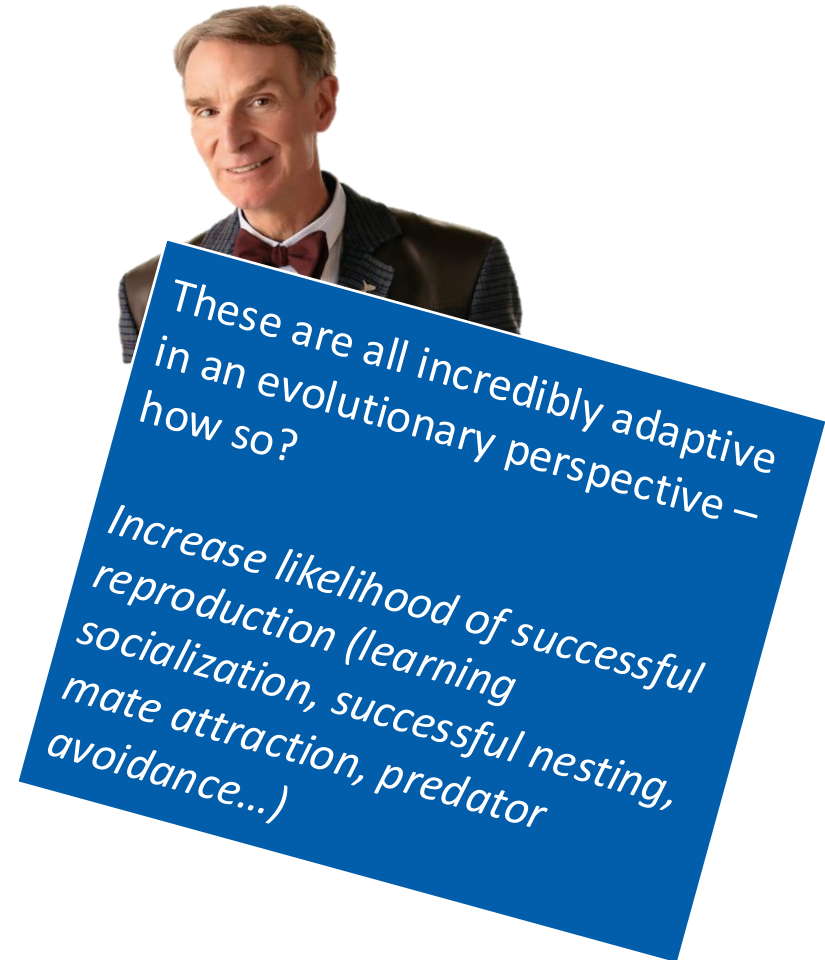
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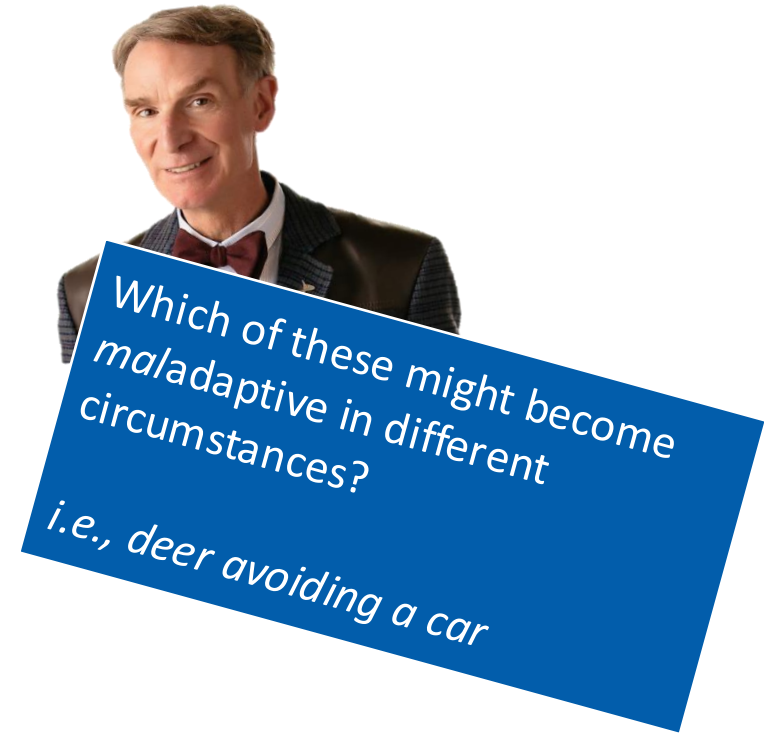


*When might these become
maladaptive?*

Elicited Behaviors

Fixed Action Patterns – fixed complex response to a stimulus

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Elicited Behaviors

Reflexes – automatic response to a stimulus

Fixed Action Patterns – fixed complex response to a stimulus

Reflexes/Fixed-Action-Patterns = Species-wide increase in chance of reproduction, but not flexible enough to adapt to a rapid change in environment (i.e., introduction of cars)

LEARNING is key

Learning – the basics

Learning

A simple, behaviorist definition of learning – when experience alters the behavioral response of some stimulus
– i.e., change in the nature of an S-R relationship

Learning

Two simple mechanisms of learning:

- Habituation – decrease strength of S-R through repeated exposure to stimulus
- Sensitization – increase strength of S-R through repeated exposure to stimulus

Learning

Habituation – decrease strength of S-R through repeated exposure to stimulus

- Unexpected tap(s) on shoulder

What kind of reflex response will an unexpected tap on the shoulder elicit?



Learning

Habituation – decrease strength of S-R through repeated exposure to stimulus

- Unexpected tap(s) on shoulder → startle response
- Followed by repeated taps

Learning

Habituation – decrease strength of S-R through repeated exposure to stimulus

- Loud knock on door → startle response
- Followed by repeated knocks

What's another example of *habituation*?



Learning

Habituation – decrease strength of S-R through repeated exposure to stimulus

- Dishabituation – sudden reappearance of response to habituated stimulus
- Loud car → startle
- Followed by other similarly loud cars
- Suddenly louder car → startle
- “re-representation of an ‘old’ stimulus as ‘new’”

Learning

- Habituation – decrease strength of S-R through repeated exposure to stimulus
- **Sensitization** – increase strength of S-R through repeated exposure to stimulus

Learning

Sensitization – increase strength of S-R through repeated exposure to stimulus

- Barking dog → anger
- Screaming child → frustration
- Each call/bell increases S-R relationship with excitement

Learning

Sensitization – increase strength of S-R through repeated exposure to stimulus

- **Can be more complex than habituation**, involving higher processes (i.e., emotions, logic, etc.)
- Repeated verbal abuse → increased aggression in response (depends on emotions, higher processing, etc.)

Learning

Sensitization – increase strength of S-R through repeated exposure to stimulus

- **Can be more generalizable than habituation**, where a *similar* stimulus can sensitize the response to the original stimulus
- Repeated exposure to fearful stimulus OR something similar → increased fear response (*PTSD, phobias, etc.*)

Learning

A simple, behaviorist definition of learning – when experience alters the behavioral response of some stimulus

Two simple mechanisms of learning:

- Habituation
- Sensitization

Both can be Short-Term (quick return to normal S-R relationship) **or Long-Term** (slow return to normal)

Learning

A simple, behaviorist definition of learning – when experience alters the behavioral response of some stimulus

Two simple mechanisms of learning:

- Habituation
- Sensitization

What dictates when an S-R is habituated vs sensitized?

- Intensity
- Adaptiveness
- Cognition

Learning

A simple, behaviorist definition of learning – when experience alters the behavioral response of some stimulus

Two simple mechanisms of learning:

- Habituation
- Sensitization

What dictates when an S-R is habituated vs sensitized?

- Intensity
 - Low-intensity usually habituates (not urgent)
 - High-intensity usually sensitizes (urgent)
- Adaptiveness
 - Rotten smells, wasp noises, barking dogs don't habituate even at lower intensities (danger)
- Cognition
 - Is the stimulus appropriate?
 - Screaming at football game vs screaming at a wedding reception (both loud, but one shouldn't have screaming)

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Two simple mechanisms of learning:

- Habituation – decrease strength of S-R through repeated exposure to stimulus
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These have been examples of learning at the behavioral level

These principles also apply at cellular level

Our *nervous systems* can learn, too!

Learning

A simple, behaviorist definition of learning is a change in behavior when experience alters the response of some stimulus.

What kind of action is this?

Two simple mechanisms of learning:

- Habituation – decrease
- Sensitization – increase

Cellular learning in the *nervous system*

Sensory
Nerve

Spine

Motor
Nerves

Spinal cord

Afferent neuron
axon (sensory)

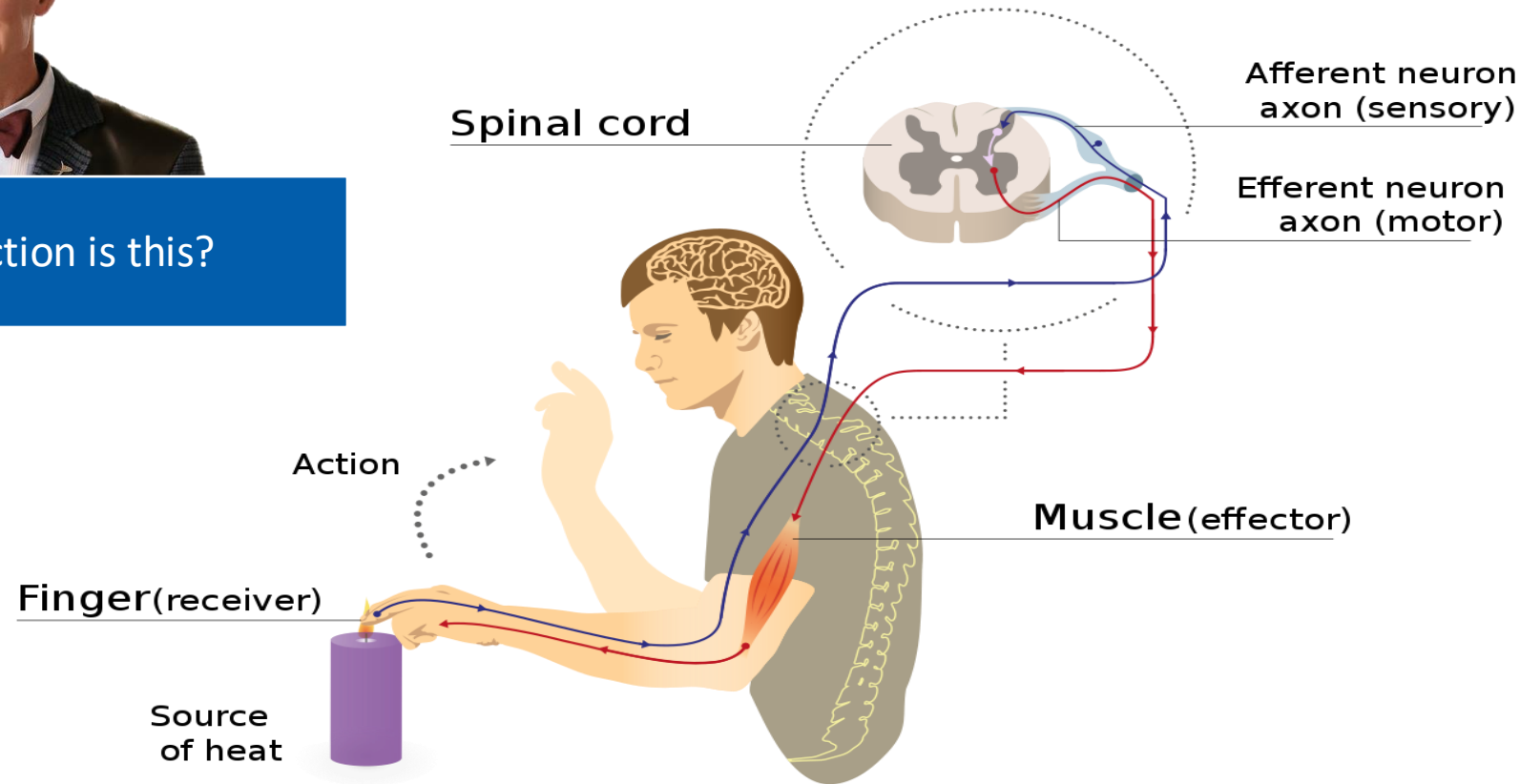
Efferent neuron
axon (motor)

Muscle (effector)

Action

Finger (receiver)

Source
of heat



Learning

A simple, behaviorist definition of learning is a change in behavior when experience alters the response of some stimulus.

What kind of action is this?
Reflex – flexion response

Two simple mechanisms of learning:

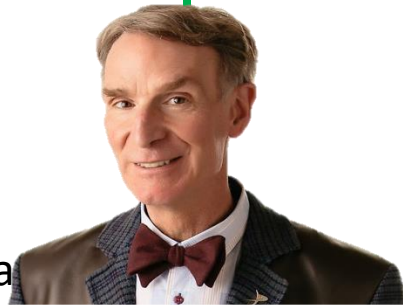
- Habituation – decrease
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Cellular learning in the *nervous system*

Sensory
Nerve

Spine

Motor
Nerves



Spinal cord

Afferent neuron
axon (sensory)

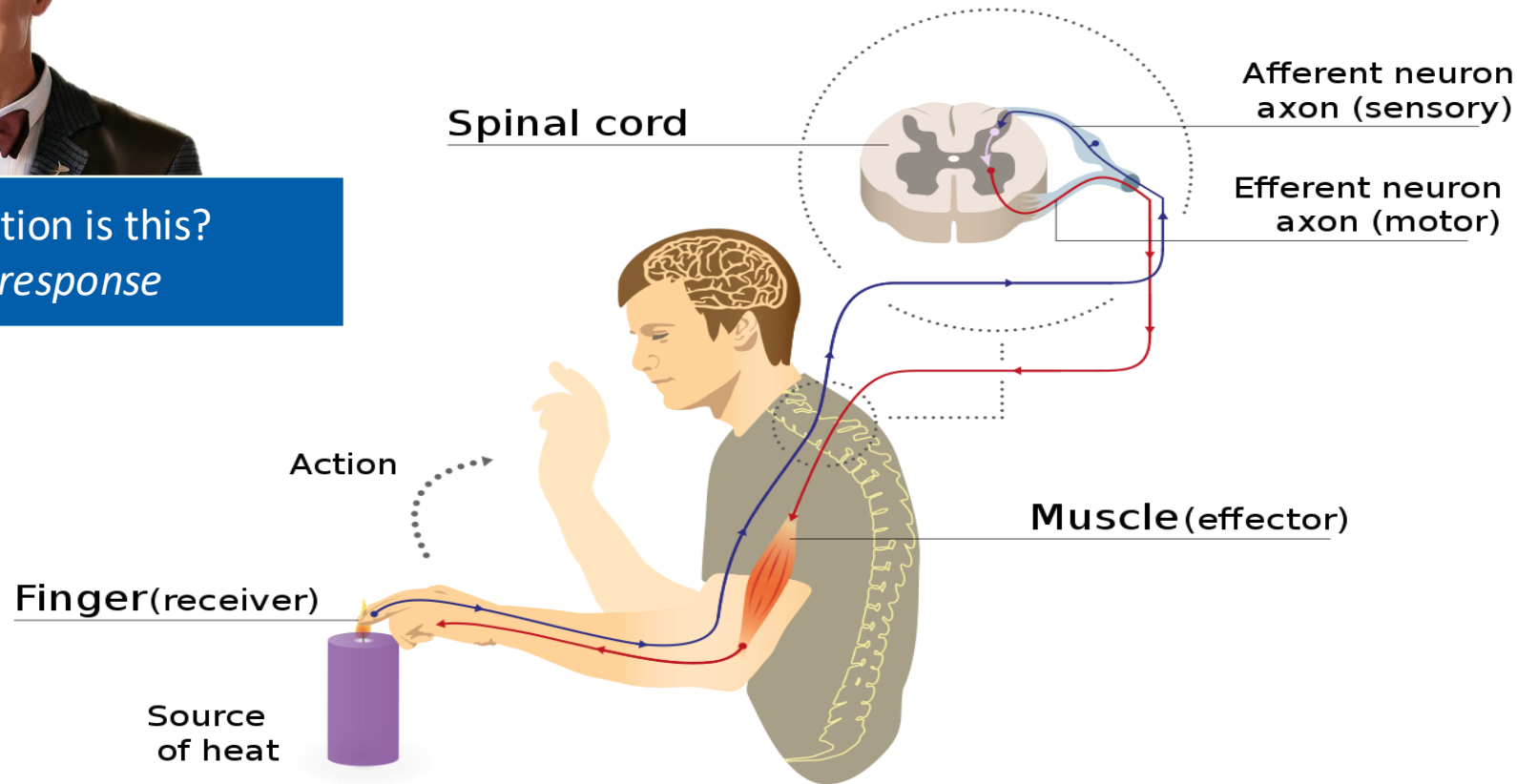
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Action

Muscle (effector)

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Source
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Two simple mechanisms of learning:

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Cellular learning in the *nervous system*

Where do you think *learning* happens in this circuit?

Sensory
Nerve

Spine

Motor
Nerves

Spinal cord

Afferent neuron
axon (sensory)

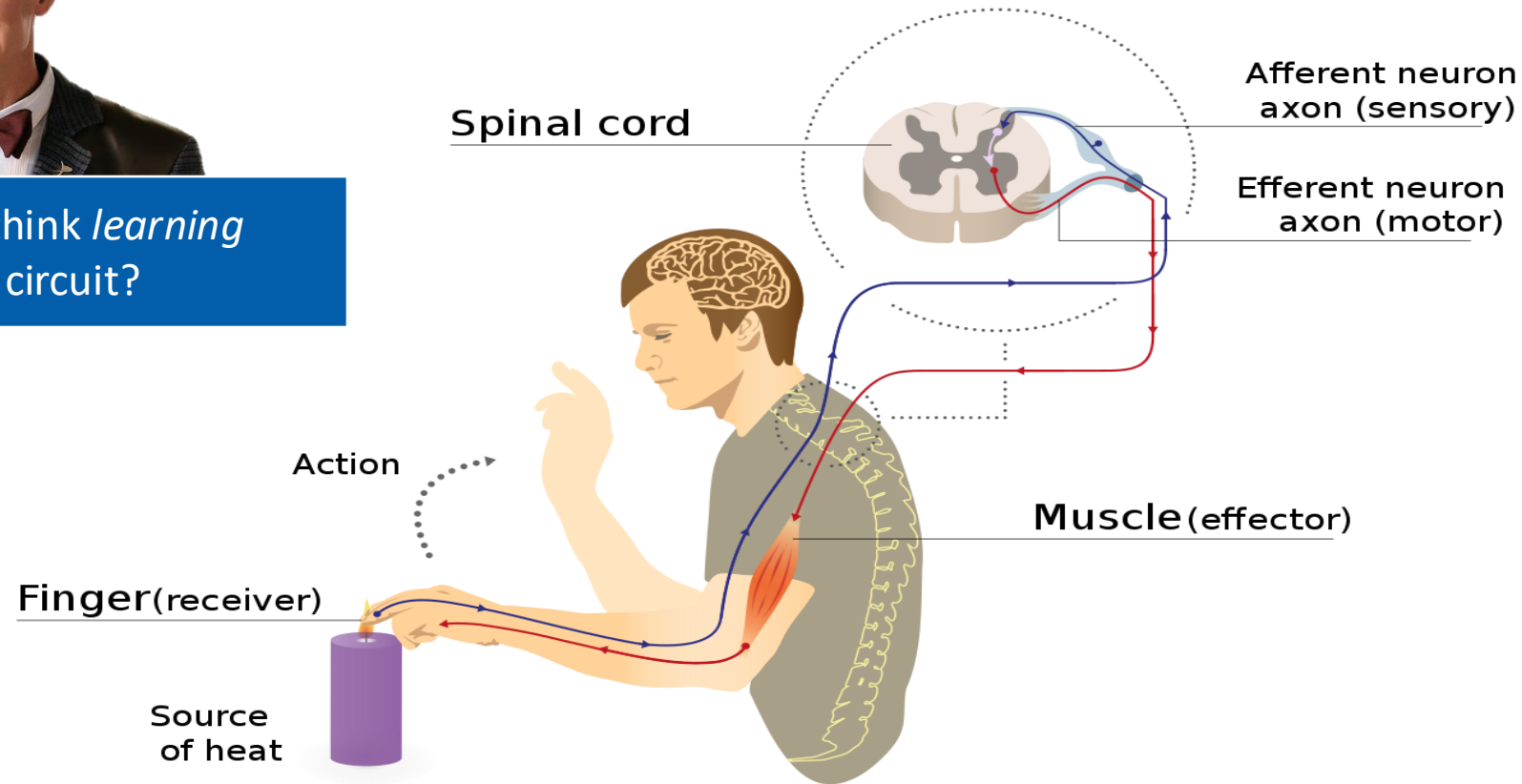
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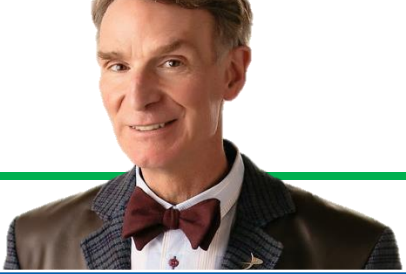
Muscle (effector)

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Source
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What dictates if the influence of Sensory on Motor is weakened or strengthened?

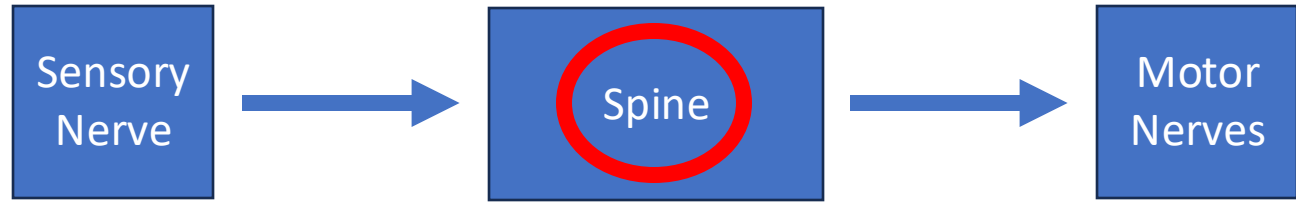
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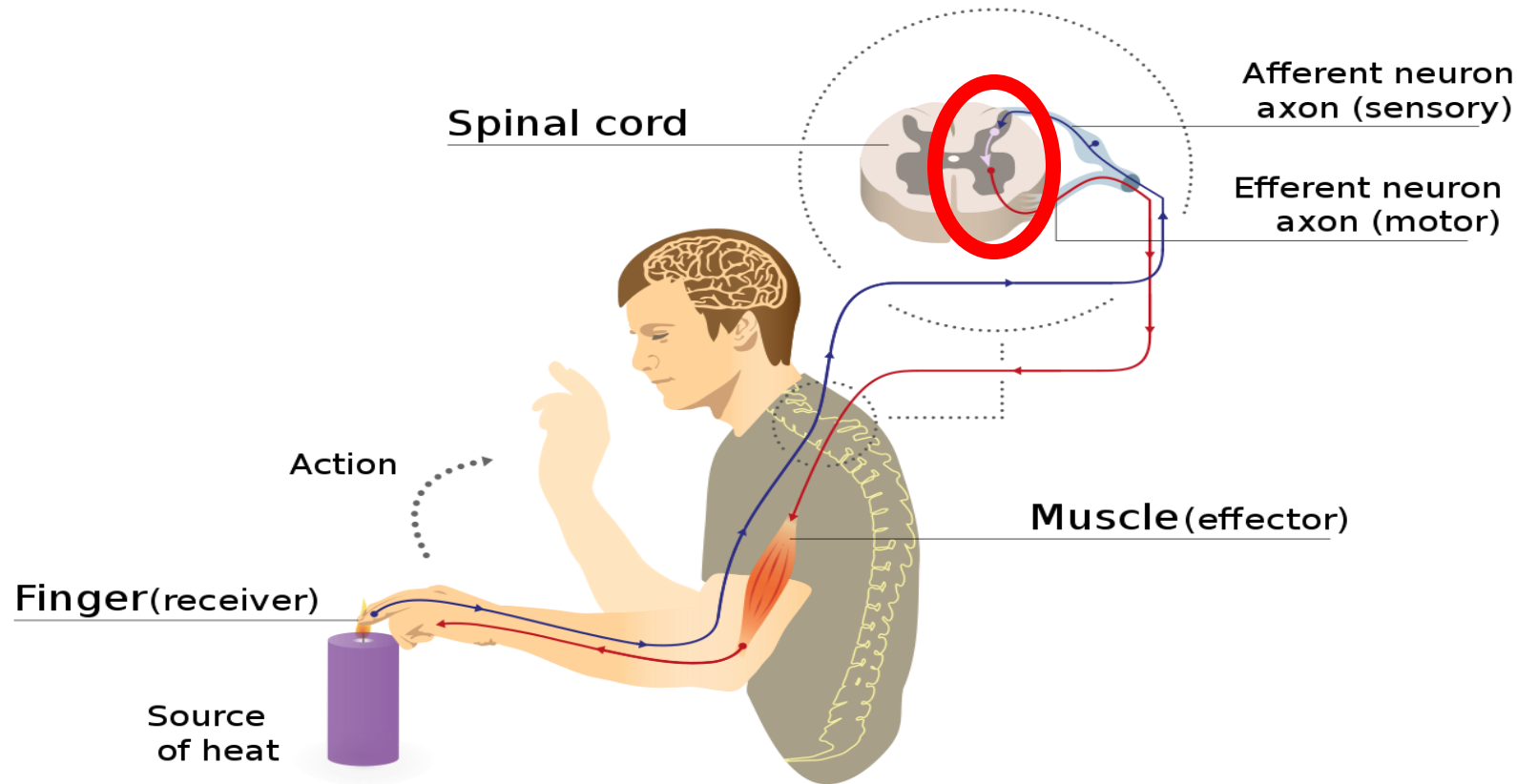
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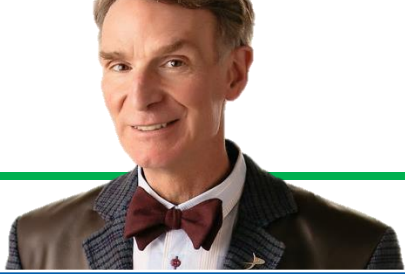
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Cellular learning in the *nervous system*



Repeated exposure → Weakened/Strengthened influence of sensory nerve on motor nerve(s)





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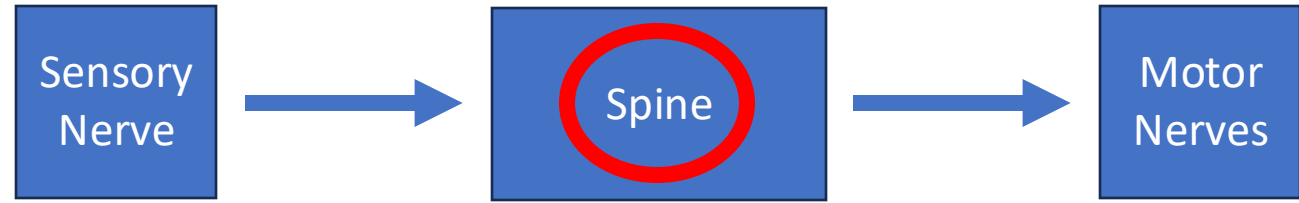
Intensity, adaptiveness, cognition

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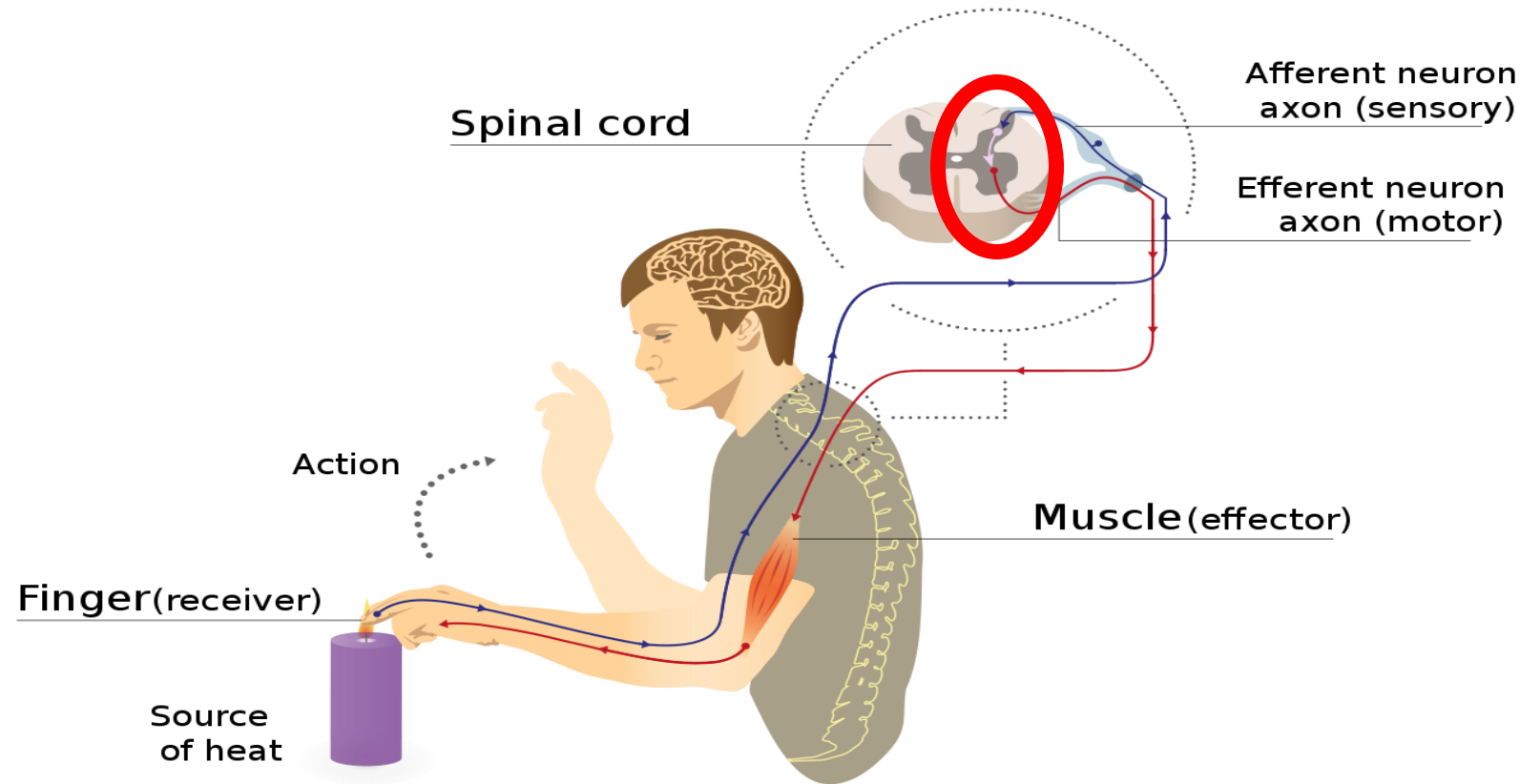
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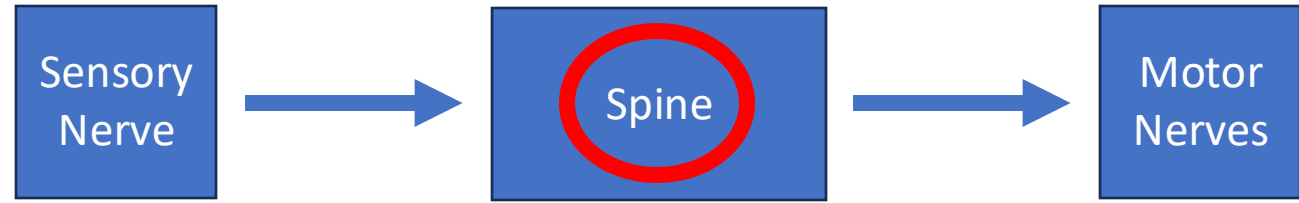
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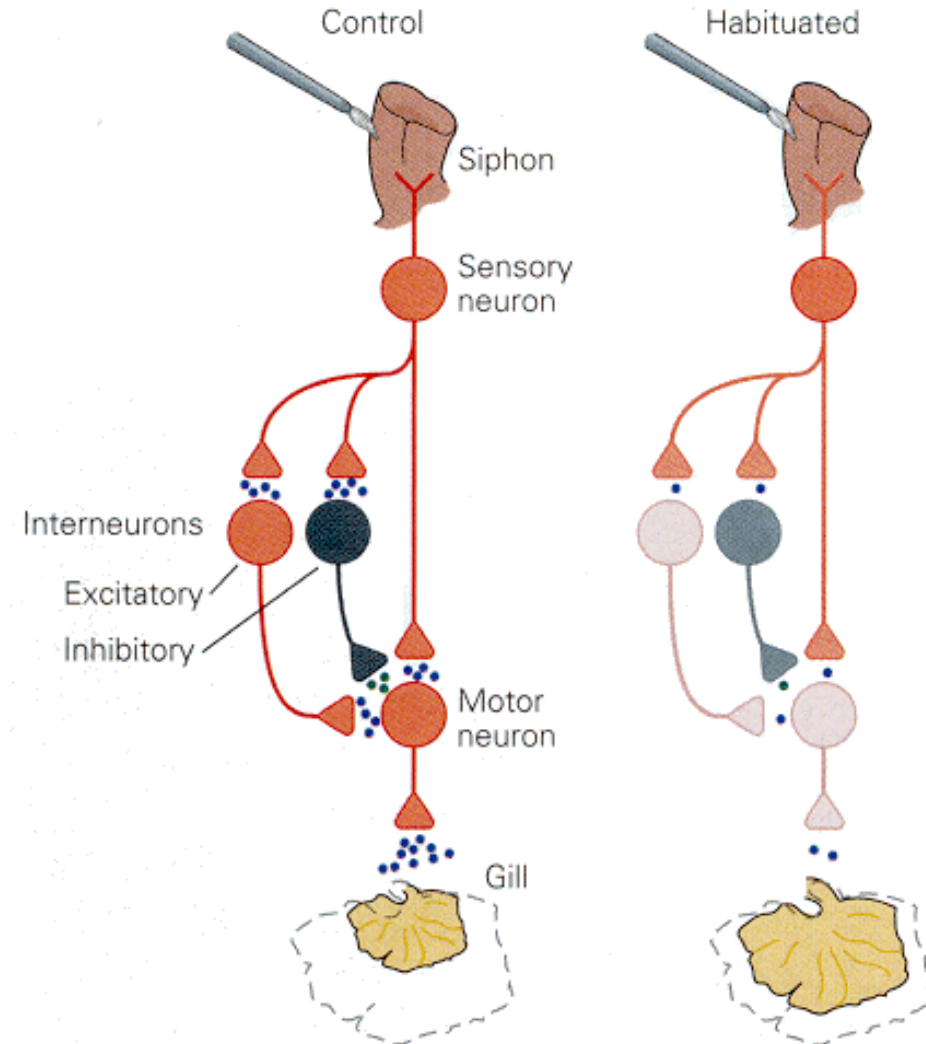
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Cellular learning in the *nervous system*

*Neurobiology of siphon-gill (S-R) withdrawal
reflex habituation (AKA, learning) in Aplysia
slugs (Nobel Prize, 2000 – Dr. Kandel)
[mentioned in book]*



Repeated exposure → Weakened/Strengthened influence of sensory nerve on motor nerve(s)



Questions?